



Ahmed TIOUAL

Data Analyst / Data Engineer

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SUMMARY

Three internships, three environments (industrial, academic, financial), two master's degrees, and one common thread: making disconnected data systems talk to each other. Reconciling a DB2 database with its AWS target following a cloud migration, replacing a manual Excel tracker with a Power BI dashboard consulted daily by quality teams, satellite classification reaching 94% accuracy. Holding a Master's Degree in Business Intelligence from the University of Lorraine and a Master's Degree in Data Science from Cadi Ayyad University, I'm looking for a Data Analyst or Data Engineer role where data reliability directly drives operational decisions.

PROFESSIONAL EXPERIENCE

Data Analyst Intern

International Electronics & Engineering (IEE)

Bissen, Luxembourg

April 2026 – Sept 2026

- Developed an end-to-end analytics platform using Python, SQL Server, and Power BI, automating data ingestion, validation, and reporting workflows.
- Collaborated closely with the Product Manager to translate requirements into scalable data models, dashboards, and automated reporting solutions.
- Replaced manual Excel reporting with a Power BI dashboard for blocked stock monitoring, improving real-time operational visibility and enabling faster incident response for quality teams.
- Built a SQL/Python data validation framework to reconcile source and target systems following a cloud migration, validating CDC replication mappings and strengthening data pipeline reliability.
- Developed PL/SQL scripts to support compliant data deletion processes and ensure adherence to established data governance policies during data management operations.

Data Scientist Intern – Remote Sensing and Climate

Center for Remote Sensing Applications (CRSA), UM6P

Benguerir, Morocco

Feb 2025 – Aug 2025

- Designed and built an end-to-end data pipeline ingesting and processing large-scale Sentinel-1/2 satellite and climate datasets via Google Earth Engine for drought monitoring.
- Engineered feature extraction and a supervised classification pipeline (XGBoost, SHAP) achieving 94% accuracy on crop type, supporting regional agricultural planning.
- Performed statistical analyses and predictive modeling (XGBoost, SHAP, SPI/SPEI) to identify relationships between climate indicators and agricultural outcomes.

Data Analyst Intern – Financial Data

Urban Agency

Guelmim, Morocco

Apr 2023 – Jun 2023

- Developed ETL pipelines integrating multiple financial data sources, significantly reducing annual reporting time.
- Digitalized financial dashboards with the finance department, enabling real-time monitoring of operational and investment KPIs and improving data reliability.

EDUCATION

- **Master's Degree in Business Intelligence** **Sept 2025 – Sept 2026**
University of Lorraine, Faculty of Mathematics, Computer Science and Mechanics – Metz, France
- **Master's Degree in Data Science** **Sept 2023 – Jul 2025**
University of Cadi Ayyad, Faculty of Sciences Semlalia – Marrakech, Morocco
- **Professional Bachelor's Degree in Statistics and Business Intelligence** **Sept 2022 – Jun 2023**
University of Ibn Zohr, Higher School of Technology – Laâyoune, Morocco

SKILLS

- **BI & Reporting:** Power BI (DAX, Power BI Desktop, Power BI Service), Tableau, Data Warehousing concepts, Dimensional Modelling, Excel (Advanced)
- **Databases:** SQL Server, Oracle DB, DB2, MySQL, BigQuery, MongoDB, Cassandra
- **Programming:** SQL, Python (Pandas, Poetry, Django, Streamlit), Java
- **Data Engineering & ETL:** ETL (IBM DataStage, Talend), IBM InfoSphere Data Replication, Spark, Kafka
- **Cloud & Tools:** Microsoft Azure, AWS, Git, GitLab, Docker, Jira, Linux & Windows
- **Data Quality:** Data Validation, Data Reconciliation, Statistical Analysis, Business Reporting
- **Languages:** English (Fluent), French (Fluent)